DCM — Dependency Continuity Module

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Authority: OOF
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Canonical Definition

Dependency Continuity Module (DCM) defines the structural conditions under which operational dependency relationships remain materially continuous, governable, traceable, and operationally stable across runtime progression, orchestration environments, delegated execution chains, and interconnected operational systems.

A system satisfies DCM only if:

- operational dependencies remain materially continuous
- dependency relationships remain traceable
- runtime dependency continuity survives execution progression
- delegated dependency chains remain operationally stable
- dependency fragmentation does not silently destabilize operational
- coordination

A system that preserves isolated execution continuity while operational dependencies materially fragment does not satisfy DCM.

Module Operational Role

DCM defines the dependency continuity layer of operational coordination governance by preserving stable operational dependency relationships across interconnected runtime environments.

Module Operational Space

DCM governs the dependency continuity space of ODCS by preserving operational dependency stability, runtime dependency traceability, delegated execution continuity, orchestration dependency continuity, and consequence-bearing dependency alignment across distributed operational systems.

Module Function

The module applies wherever systems must preserve:

- operational dependency continuity
- runtime dependency stability
- orchestration dependency continuity
- delegated execution dependency continuity
- distributed dependency traceability
- consequence-bearing dependency stability

Its function is to ensure that operational dependency relationships remain materially stable strongly enough to preserve coordination-valid operational continuity.

Minimum Implementation Framework

1. Define the Dependency Object

The organization must define which operational dependencies require continuity preservation.

This may include:

- orchestration dependencies
- runtime dependencies
- delegated execution dependencies
- coordination chains
- operational coupling states
- synchronization dependencies
- consequence-bearing dependency relationships

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2. Define Dependency Continuity Conditions

The system must define the conditions under which operational dependencies remain materially continuous.

This includes:

- runtime dependency continuity
- orchestration continuity
- delegated coordination continuity
- operational coupling stability
- dependency traceability continuity

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3. Define Dependency Fragmentation Detection Logic

The system must define how dependency fragmentation or unstable dependency continuity is identified.

This may include:

- broken coordination chains
- unstable runtime dependencies
- delegated execution fragmentation
- hidden dependency drift
- consequence-bearing dependency instability

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4. Define Operational Response or Governance Logic

The system must define governance logic for materially unstable dependency continuity conditions.

Governance response may include:

- dependency escalation
- orchestration restriction
- runtime narrowing
- coordination review activation
- dependency stabilization enforcement
- operational invalidation where required

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5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of dependency continuity and dependency-fragmentation states. A system must not remain coordination-valid if operational dependencies materially fragment while systems continue assuming stable coordination continuity remains preserved.

Use Case 1 — AI Orchestration Dependency

Chain

Scenario

A distributed AI environment coordinates runtime execution across multiple orchestration systems, delegated agents, and dependency-bound operational chains.

Application

DCM preserves stable dependency continuity across orchestration progression and delegated execution environments.

Result

The environment gains stronger operational coordination stability and reduced dependency fragmentation across AI runtime systems.

Use Case 2 — Enterprise Operational Workflow

Scenario

An enterprise operational environment depends on coordinated execution across multiple interconnected systems and runtime-dependent operational chains.

Application

DCM preserves operational dependency continuity across distributed workflow execution environments.

Result

The organization gains stronger coordination resilience and reduced dependency instability across operational systems.

Canonical Closing Statement

If operational dependencies cannot remain materially continuous across interconnected environments, systems may preserve local execution continuity while operational coordination silently fragments underneath.

Related Documents

→ Operational Dependency & Coordination Standard - (ODCS)

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